

 [_ \(https://twitter.com/ai4k12\)](https://twitter.com/ai4k12)

 [_ \(https://www.youtube.com/channel/UCsEdLohr2U6uW6lQQ9ovBrw\)](https://www.youtube.com/channel/UCsEdLohr2U6uW6lQQ9ovBrw)



(<https://ai4k12.org/>)

List of Resources

Card View
(<https://ai4k12.org/resources-card-view/>)

AI-Powered Educational Robots

- AI-Powered Educational Robot: AutoAuto from Master AI (<https://ai4k12.org/autoauto-from-master-ai/>)
- AI-Powered Educational Robot: mBot2 from MakeBlock (<https://ai4k12.org/mbot2-from-makeblock/>)
- AI-Powered Educational Robot: VEX AIM from Innovation First (<https://ai4k12.org/vex-aim-coding-robot/>)

Books

- Book: AI Super-Powers: China, Silicon Valley, and The New World Order (<https://ai4k12.org/book-ai-super-powers-china-silicon-valley-and-the-new-world-order/>)
- Book: Architects of Intelligence: The truth about AI from the people building it (<https://ai4k12.org/book-architects-of-intelligence-the-truth-about-ai-from-the-people-building-it/>)
- Book: Artificial Intelligence: A Guide for Thinking Humans (<https://ai4k12.org/artificial-intelligence-a-guide-for-thinking-humans/>)

- Book: Artificial Intelligence: A Modern Approach (<https://ai4k12.org/book-artificial-intelligence-a-modern-approach/>)
- Book: How Smart Machines Think (<https://ai4k12.org/how-smart-machines-think/>)
- Book: How to Teach AI: Weaving Strategies and Activities Into Any Content Area (<https://ai4k12.org/how-to-teach-ai-weaving-strategies-and-activities-into-any-content-area/>)
- Book: Teaching AI: Exploring New Frontiers for Learning (<https://ai4k12.org/teaching-ai-exploring-new-frontiers-for-learning/>)
- Book: The Future Computed: Artificial Intelligence and Its Role in Society (<https://ai4k12.org/the-future-computed-artificial-intelligence-and-its-role-in-society/>)
- Book: The Learner's Apprentice: AI and the Amplification of Human Creativity (<https://ai4k12.org/the-learners-apprentice-ai-and-the-amplification-of-human-creativity/>)
- Book: What to Think About Machines That Think (<https://ai4k12.org/what-to-think-about-machines-that-think/>)
- Book Chapter: Artificial Intelligence Thinking in K-12 (<https://ai4k12.org/artificial-intelligence-thinking-in-k-12/>)
- Book Series: Hands-On AI Projects for the Classroom (<https://ai4k12.org/hands-on-ai-projects-for-the-classroom/>)
- Book: Power On (<https://ai4k12.org/power-on/>)

Reports

- Report: Automation and Artificial Intelligence: How machines are affecting people and places (<https://ai4k12.org/report-automation-and-artificial-intelligence-how-machines-are-affecting-people-and-places/>)
- Report: Building Capacity for K-12 Artificial Intelligence Education Research (<https://ai4k12.org/building-capacity-for-k-12-artificialintelligence-education-research/>)
- Report Executive Summary: State of AI Education in Your State (<https://ai4k12.org/executive-summarythe-state-of-k-12-ai-education-in-your-state-workshop/>)
- Report: The Future of Problem Solving with Data and Intelligence: Increasing Artificial Intelligence and Data Science Education Across the US (<https://ai4k12.org/the-future-of-problem-solving-with-data-and-intelligence-increasing-artificial-intelligence-and-data-science-education-across-the-us/>)
- Report: K-8 AI Career Competencies (http://stelar.edc.org/sites/stelar.edc.org/files/K-8_STEM_Career_Competencies.pdf) (site under maintenance)

Careers

- Career Training: Robotics Career (<https://ai4k12.org/robotics-career/>)

- Career Training: AI4GA's AI Career Cards (<https://ai4k12.org/ai4gas-ai-career-cards/>)

Children's Books

- Book: A Fresh Squeeze on Data (<https://ai4k12.org/book-a-fresh-squeeze-on-data/>)
- Book: How to Train Your Robot (<https://ai4k12.org/how-to-train-your-robot/>)
- Book Series: AI and ME (ReadyAI) (<https://ai4k12.org/book-series-ai-and-me/>)
- Book Series: AI Made Simple (edsquare) (<https://ai4k12.org/ai-made-simple-ed-square/>)
- Book Series: The World of Artificial Intelligence 4D (<https://ai4k12.org/the-world-of-artificial-intelligence-4d/>)
- Book Series: Tinker Toddlers (<https://ai4k12.org/book-series-tinker-toddlers/>)

Competitions

- Competition: Halite (<https://ai4k12.org/competition-halite/>)
- Competition: WAICY (World Artificial Intelligence Competition for Youth) (<https://ai4k12.org/waicy-world-artificial-intelligence-competition-for-youth/>)

CTE/CTAE Pathways

- CTAE Pathways: Denmark High School (Georgia) Artificial Intelligence Pathway (<https://ai4k12.org/denmark-high-school-georgia-artificial-intelligence-pathway/>)
- CTE Pathways: Walker Junior High and Kennedy High School (California) Artificial Intelligence Pathway (<https://ai4k12.org/walker-junior-high-and-kennedy-high-school-california-artificial-intelligence-pathway/>)
- Georgia's AI CTAE Frameworks (Approved) – Artificial Intelligence Pathway (<https://ai4k12.org/georgias-ai-ctae-frameworks/>)
 - Foundations of Artificial Intelligence
 - Artificial Intelligence Concepts
 - Artificial Intelligence Applications
- Florida's AI CTE Framework (Under Review) – Draft 4 Course Sequence – (<https://ai4k12.org/floridas-ai-cte-framework/>) AI in the World, Applications of AI, Procedural Program (Python), Foundations of Machine Learning

Curriculum Materials

- Curriculum: AI4ALL Open Learning (<https://ai4k12.org/curriculum-ai4all-open-learning/>)
- Curriculum: AI & Cybersecurity for Teens (ACT) (<https://ai4k12.org/ai-cybersecurity-for->)

teens-act/)

- Curriculum: AI and Machine Learning Module (Code.org middle school curriculum) (<https://ai4k12.org/ai-and-machine-learning-module/>)
- Curriculum: AI for Georgia (AI4GA) (middle school curriculum) (<https://ai4k12.org/ai-for-georgia-ai4ga/>)
- Curriculum: AI for Oceans (Code.org machine learning intro) (<https://ai4k12.org/ai-for-oceans/>)
- Curriculum: AI plus Ethics for Middle School (Blakely Payne, MIT) (<https://ai4k12.org/ai-plus-ethics-for-middle-school/>)
- Curriculum: Code.org AI Curricula (Code.org) (<https://ai4k12.org/code-org-ai-curricula/>)
- Curriculum: Computational Thinking & AI in Elementary School (Claude Reuter) (<https://ai4k12.org/computational-thinking-ai-in-elementary-school/>)
- Curriculum: CRAFT (Classroom-Ready Resources About AI for Teaching) [Stanford] (<https://ai4k12.org/craft-classroom-ready-resources-about-ai-for-teaching/>)
- Curriculum: CS Frontiers: AI/ML (<https://ai4k12.org/cs-frontiers-ai-ml/>)
- Curriculum: DAILY Curriculum (Irene Lee, MIT) (<https://ai4k12.org/daily-curriculum/>)
- Curriculum: ECS Alternate Curriculum Unit on Artificial Intelligence (<https://ai4k12.org/ecs-alternate-curriculum-unit-on-artificial-intelligence/>)
- Curriculum: Exploring the Future Through Robotics and Artificial Intelligence (Matthew Reuben) (<https://ai4k12.org/exploring-the-future-through-robotics-and-artificial-intelligence/>)
- Curriculum: FarmBeats for Students (Microsoft Education) (<https://ai4k12.org/farm-beats-for-students/>)
- Curriculum: IBM Activity Kits on AI (<https://ai4k12.org/ibm-activity-kits-on-ai/>)
- Curriculum: Thinking with KIBO: Introducing Artificial Intelligence in Early Grades (KinderLab Robotics) (<https://ai4k12.org/thinking-with-kibo-introducing-artificial-intelligence-in-early-grades/>)

Demos

- Demos: Convolutional Neural Nets (<https://ai4k12.org/demos-convolutional-neural-nets/>)
- Demo: ConvNetJS Convolutional neural net demos (<https://ai4k12.org/convnetjs/>)
- Demo: Diffusion Demo (<https://ai4k12.org/diffusion-demo/>)
- Demo: Erase Your Face (<https://ai4k12.org/erase-your-face/>)
- Demo: FaceDemo: TinyYOLOV2 Face Detection (<https://ai4k12.org/facedemo-tiny-yolo2-face-detection/>)
- Demo: FaceMesh: Real-time face mesh from webcam (<https://ai4k12.org/facemesh/>)

- Demo: Hand Pose Detector: Real-time palm and finger detection from webcam images (<https://ai4k12.org/hand-pose-detector/>)
- Demo: Markov Chain Demo: Generate text using a statistical language model (<https://ai4k12.org/markov-chain-demo/>)
- Demo: Neuron Sandbox: Solve problems using a simulated linear threshold unit. (<https://ai4k12.org/neuron-sandbox/>)
- Demo: Object Detection: Real-time object detection from webcam images (<https://ai4k12.org/object-detection/>)
- Demo: pix2pix (<https://ai4k12.org/pix2pix/>)
- Demo: Quick, Draw! (<https://ai4k12.org/quick-draw/>)
- Demo: Scroobly Doodle Animation (<https://ai4k12.org/scroobly/>)(video (<https://www.youtube.com/watch?v=mq2jK5X9JNY>))
- Demos: Spectrograms and Waveforms (<https://ai4k12.org/spectrograms-and-waveforms/>)
- Demo: Speech Recognition in Chrome (<https://ai4k12.org/speech-recognition-in-chrome/>)
- Demo: Teachable Machine (<https://ai4k12.org/teachable-machine/>)
- Demo: Extension of Teachable Machine (GenAI Project, Finland) (<https://ai4k12.org/extension-of-teachable-machine/>)
- Demo: TensorFlow Playground (<https://ai4k12.org/tensorflow-playground/>)
- Demo: Word Embedding Demo (<https://ai4k12.org/word-embedding-demo/>)
- Demo: TensorFlow.js Models (<https://ai4k12.org/tensorflow-js-models/>)(video (<https://www.youtube.com/watch?v=c3aZeMZ0cvs>))
- Demo: University of Karlsruhe (<https://lecture-demo.ira.uka.de/>)

Lists of Links

- Links: DeepFakes (<https://ai4k12.org/links-deepfakes/>)
- Links: Diversity and Inclusion in AI (<https://ai4k12.org/diversity-and-inclusion-in-ai/>)
- Links: Transformer Networks (<https://ai4k12.org/transformer-networks/>)

Online Professional Development Courses

- Online Course: AI 101 for Teachers (Code.org) (<https://ai4k12.org/ai-101-for-teachers/>)
- Online Course: Artificial Intelligence Explorations and Their Practical Use in Schools (ISTE) (<https://ai4k12.org/online-course-artificial-intelligence-explorations-and-their-practical-use-in-schools/>)

- Online Course: IBM AI Education (IBM/MindSpark) (<https://ai4k12.org/ibm-ai-education/>)
- Online Course: Teaching AI in the Classroom (CSER, Australia) (<https://ai4k12.org/teaching-ai-in-the-classroom/>)
- Online Course: Artificial Intelligence (AI) Education for Teachers (coursera) (<https://ai4k12.org/artificial-intelligence-ai-education-for-teachers/>)

Online Courses for K-12 Students

- Online Course: AI and ME (ReadyAI) (<https://ai4k12.org/online-course-ai-and-me/>)
- Online Course: AI Foundations Powered by ISTE and IBM (<https://ai4k12.org/ai-foundations-powered-by-iste-and-ibm/>)
- Online Course: AI4All Open Learning (AI4All) (<https://ai4k12.org/ai4all-open-learning/>)
- Online Course: Elements of AI (Reaktor + University of Helsinki) (<https://ai4k12.org/elements-of-ai/>)
- Online Course: Introduction to Q-learning in Python (LoisLab) (<https://ai4k12.org/introduction-to-q-learning-in-python/>)

Reference Sources and Tutorials

- The Artificial Intelligence Wiki (<https://ai4k12.org/the-artificial-intelligence-wiki/>) (Path-Mind): encyclopedia-like accessible explanations of concepts in AI reasoning and machine learning
- Towards Data Science (<https://ai4k12.org/towards-data-science/>): blog with a mix of introductory articles and detailed technical explorations
- An Explanation of Machine Learning Even You Can Understand (<https://ai4k12.org/an-explanation-of-machine-learning-even-you-can-understand/>) (TheNextWeb): quick survey of the fundamentals
- The difference between AI and machine learning, explained (<https://ai4k12.org/the-difference-between-ai-and-machine-learning-explained/>)(TheNextWeb): insightful discussion of the nature of AI and its relationship to ML

Research

- NSF AI Education for K-12 Awards (<https://ai4k12.org/nsf-ai-education-for-k-12-awards-fy18-fy21/>)
- EAAI K-12 AI Education Papers (<https://ai4k12.org/eaai-k-12-ai-education-papers/>)

Resource Directories

- Resource Directory: Expore AI Ethics (<https://ai4k12.org/explore-ai-ethics/>)
- Resource Directory: Learn About AI at Code.org (<https://ai4k12.org/learn-about-ai-at-code-org/>)
- Resource Directory: MIT RAISE (Responsible AI for Social Empowerment and Education) (<https://ai4k12.org/raise-playground/>)
- The Best Online Machine Learning Courses (2022) (<https://ai4k12.org/the-best-online-machine-learning-courses-2022/>)

Software Packages

- Software: AI and MIT App Inventor (<https://ai4k12.org/software-ai-with-mit-app-inventor/>)
- Software: Calypso (intelligent robot programming framework) (<https://ai4k12.org/calypso/>)
- Software: Cognimates (<https://ai4k12.org/cognimates/>)
- Software: eCraft2Learn (<https://ai4k12.org/ecraft2learn/>)
- Software: Machine Learning for Kids (<https://ai4k12.org/machine-learning-for-kids/>)
- Software: micro:bit CreateAI (<https://ai4k12.org/microbit-createai/>) and micro:bit ML-Machine (<https://ai4k12.org/microbit-ml-machine/>)
- Software: RAISE Playground (<https://ai4k12.org/raise-playground/>)

TeachAI Resources

- TeachAI Resources: AI Guidance for Schools Toolkit (2023) (<https://ai4k12.org/ai-guidance-for-schools-toolkit/>)
- TeachAI Resources: Future of Computer Science Education in an Age of AI (2024) (<https://ai4k12.org/future-of-computer-science-education-in-an-age-of-ai-2024/>)
- TeachAI Resources: Foundational Policy Ideas for AI in Education (2024) (<https://ai4k12.org/foundational-policy-ideas-for-ai-in-education-2024/>)

Videos

- Documentary: AlphaGo (<https://ai4k12.org/documentary-alphago/>)
- Documentary: CyberWork and the American Dream (<https://ai4k12.org/cyberwork-and-the-american-dream/>)
- Documentary: Frontline: In the Age of AI (<https://ai4k12.org/frontline-in-the-age-of-ai/>)
- Documentary: Nova: Look Who's Driving (<https://ai4k12.org/nova-look-whos-driving/>)
- Video Playlist: Computer Vision (5 videos on computer vision projects at Google) (<https://ai4k12.org/playlist-computer-vision/>)

- Video Series: Careers in AI (<https://ai4k12.org/series-careers-in-ai/>)
- Video Series: Crash Course Artificial Intelligence (20 episodes) (<https://ai4k12.org/crash-course-artificial-intelligence/>)
- Video Series: Crash Course Computer Science (5 episodes deal with AI) (<https://ai4k12.org/crash-course-computer-science/>)
- Video Series: EverydAI (<https://ai4k12.org/everydai/>)
- Video Series: How AI Works (from Code.org) (<https://ai4k12.org/how-ai-works/>)
- Video Series: The Age of AI (YouTube Learning; 8 episodes) (<https://ai4k12.org/the-age-of-ai/>)
- Video: AI Ethics and Fairness in 7 Minutes (<https://ai4k12.org/ai-ethics-and-fairness-in-7-minutes/>)
- Video: Computer Scientist Explains Machine Learning in 5 Levels of Difficulty (<https://ai4k12.org/computer-scientist-explains-machine-learning-in-5-levels-of-difficulty/>)
- Video: Deconstructing Robotics (<https://ai4k12.org/deconstructing-robots/>)
- Video: How Computer Vision is Finally Taking Off, After 50 Years (<https://ai4k12.org/how-computer-vision-is-finally-taking-off-after-50-years/>)
- Video: How Computer Vision Works (<https://ai4k12.org/how-computer-vision-works/>)
- Videos: How Self-Driving Cars Work (<https://ai4k12.org/how-self-driving-cars-work/>)
- Video: How Snapchat's Filters Work (<https://ai4k12.org/how-snapchats-filters-work/>)
- Video: How to Train Your Robot (for K-8) (<https://ai4k12.org/how-to-train-your-robot-2/>)
- Video: Humans Need Not Apply (<https://ai4k12.org/humans-need-not-apply/>)
- Video: Inside Google Translate (<https://ai4k12.org/inside-google-translate/>)
- Video: Lidar vs. Vslam For Robot Vacuums (<https://ai4k12.org/lidar-vs-vslam-for-robot-vacuums/>)
- Video: Neural Networks for Babies (<https://ai4k12.org/neural-networks-for-babies/>)
- Video: The Companion Robot Designer (Bloomberg News) (<https://ai4k12.org/the-companion-robot-designer/>)
- Video: The Power and Limits of Artificial Intelligence (<https://ai4k12.org/the-power-and-limits-of-artificial-intelligence/>)
- Video: The Rise of AI (<https://ai4k12.org/the-rise-of-ai/>)



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